

# Pre-requisites

## Formulae

# Area and Volume

**Circle** : Area =  $\pi r^2$ ; Circumference =  $2\pi r$ .

**Square** : Area =  $x^2$ ; Perimeter =  $4x$ .

**Rectangle**: Area =  $xy$ ; Perimeter =  $2(x+y)$ .

**Triangle** : Area =  $\frac{1}{2}$  (base)(height); Perimeter =  $a+b+c$ .

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} a^2.$$

**Sphere** : Surface Area =  $4\pi r^2$ ; Volume =  $\frac{4}{3}\pi r^3$ .

**Cube** : Surface Area =  $6a^2$ ; Volume =  $a^3$ .

**Cone** : Curved Surface Area =  $\pi rl$ ; Volume =  $\frac{1}{3}\pi r^2 h$

$$\text{Total surface area} = \pi r l + \pi r^2$$

**Cuboid** : Total surface area =  $2(ab + bh + lh)$ ; Volume =  $lbh$ .

**Cylinder** : Curved surface area =  $2\pi rh$ ; Volume =  $\pi r^2 h$

$$\text{Total surface area (open)} = 2\pi rh;$$

$$\text{Total surface area (closed)} = 2\pi rh + 2\pi r^2.$$

# Algebraic Formulae

$$1.(a + b)^2 = a^2 + 2ab + b^2 .$$

$$2. (a - b)^2 = a^2 - 2ab + b^2 .$$

$$3.(a + b)^3 = a^3 + b^3 + 3ab(a + b).$$

$$4. (a - b)^3 = a^3 - b^3 - 3ab(a - b).$$

$$5.(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca.$$

$$6.(a + b + c)^3 = a^3 + b^3 + c^3 + 3a^2b + 3a^2c + 3b^2c + 3b^2a + 3c^2a + 3c^2b + 6abc.$$

$$7.a^2 - b^2 = (a + b)(a - b) .$$

$$8.a^3 - b^3 = (a - b) (a^2 + ab + b^2) .$$

$$9.a^3 + b^3 = (a + b) (a^2 - ab + b^2) .$$

$$10.(a + b)^2 + (a - b)^2 = 4ab.$$

$$11.(a + b)^2 - (a - b)^2 = 2(a^2 + b^2) .$$

$$12.If a + b + c = 0, then a^3 + b^3 + c^3 = 3 abc .$$

# Indices Formulae

$$1. a^m a^n = a^{m+n} \quad 2. \frac{a^m}{a^n} = a^{m-n} \quad 3. (a^m)^n = a^{mn} \quad 4. (ab)^m = a^m b^m$$

$$5. \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad 6. a^0 = 1, \quad a \neq 0 \quad 7. a^{-m} = \frac{1}{a^m} \quad 8. a^x = a^y \Rightarrow x = y$$

$$9. a^x = b^x \Rightarrow a = b \quad 10. \sqrt{a \pm 2\sqrt{b}} = \sqrt{x} \pm \sqrt{y}, \text{ where } x + y = a \text{ and } xy = b.$$

# Trigonometry

$$\text{Area of a sector of a circle} = \frac{1}{2}r^2\theta.$$

$$\text{Arc length, } S = r\theta.$$

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}, \cos\theta = \frac{\text{adj}}{\text{hyp}}, \tan\theta = \frac{\text{opp}}{\text{adj}}, \cot\theta = \frac{\text{adj}}{\text{opp}}, \sec\theta = \frac{\text{hyp}}{\text{adj}}, \text{cosec}\theta = \frac{\text{hyp}}{\text{opp}}.$$

$$\sin\theta = \frac{1}{\text{cosec}\theta} \text{ or } \text{cosec}\theta = \frac{1}{\sin\theta}, \cos\theta = \frac{1}{\sec\theta} \text{ or } \sec\theta = \frac{1}{\cos\theta},$$

$$\tan\theta = \frac{1}{\cot\theta} \text{ or } \cot\theta = \frac{1}{\tan\theta}, \tan\theta = \frac{\sin\theta}{\cos\theta}, \cot\theta = \frac{\cos\theta}{\sin\theta}.$$

$$\sin^2\theta + \cos^2\theta = 1; \Rightarrow \sin^2\theta = 1 - \cos^2\theta; \cos^2\theta = 1 - \sin^2\theta;$$

$$\sec^2\theta - \tan^2\theta = 1; \Rightarrow \sec^2\theta = 1 + \tan^2\theta; \tan^2\theta = \sec^2\theta - 1;$$

$$\text{cosec}^2\theta - \cot^2\theta = 1; \Rightarrow \text{cosec}^2\theta = 1 + \cot^2\theta; \cot^2\theta = \text{cosec}^2\theta - 1.$$

# Standard Angles

|       | $0^0$ or $0$ | $30^0$ or $\frac{\pi}{6}$ | $45^0$ or $\frac{\pi}{4}$ | $60^0$ or $\frac{\pi}{3}$ | $90^0$ or $\frac{\pi}{2}$ | $15^0$ or $\frac{\pi}{12}$      | $75^0$ or $\frac{5\pi}{12}$     |
|-------|--------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------------|---------------------------------|
| Sin   | 0            | $\frac{1}{2}$             | $\frac{1}{\sqrt{2}}$      | $\frac{\sqrt{3}}{2}$      | 1                         | $\frac{\sqrt{3}-1}{2\sqrt{2}}$  | $\frac{\sqrt{3}+1}{2\sqrt{2}}$  |
| Cos   | 1            | $\frac{\sqrt{3}}{2}$      | $\frac{1}{\sqrt{2}}$      | $\frac{1}{2}$             | 0                         | $\frac{\sqrt{3}+1}{2\sqrt{2}}$  | $\frac{\sqrt{3}-1}{2\sqrt{2}}$  |
| Tan   | 0            | $\frac{1}{\sqrt{3}}$      | 1                         | $\sqrt{3}$                | $\infty$                  | $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ | $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ |
| Cot   | $\infty$     | $\sqrt{3}$                | 1                         | $\frac{1}{\sqrt{3}}$      | 0                         | $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ | $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ |
| Sec   | 1            | $\frac{2}{\sqrt{3}}$      | $\sqrt{2}$                | 1                         | $\infty$                  | $\frac{2\sqrt{2}}{\sqrt{3}+1}$  | $\frac{2\sqrt{2}}{\sqrt{3}-1}$  |
| Cosec | $\infty$     | 2                         | $\sqrt{2}$                | $\frac{2}{\sqrt{3}}$      | 1                         | $\frac{2\sqrt{2}}{\sqrt{3}-1}$  | $\frac{2\sqrt{2}}{\sqrt{3}+1}$  |